Dataset Integrity & Target Leakage Audit Report

Project: AI-Powered Crop Pest Prediction | Date: 2026-10-05 | Lead ML Engineer

1. Executive Summary & Audit Scope

A forensic technical audit of the master dataset (249,700 rows, 38 raw columns) was executed to detect target leakage, temporal post-harvest dependencies, and synthetic redundancies. Zero data leakage was verified in the production feature matrix.

2. Column Elimination Matrix (Leakage Removal)

Column Name	Category	Action	Justification
Outbreak	Target Leakage	REMOVED	Deterministic binary proxy for Risk_Level
Risk_Score	Target Leakage	REMOVED	Formulaic linear combination of target classes
Yield	Post-Harvest Leakage	REMOVED	Measured post-harvest; unknown at pest scouting
Production_Tons	Post-Harvest Leakage	REMOVED	Harvest volume variable causing temporal leakage
Area_Hectares	Farm Economic	REMOVED	Non-climatic acreage variable
Crop_Health_Index	Post-Harvest Leakage	REMOVED	Derived from Yield / Mean_Yield
Recommended_Pesticide	Advisory Output	ISOLATED	Generated strictly post-prediction via KB
Recommended_Fertilizer	Advisory Output	ISOLATED	Generated strictly post-prediction via KB
Chemical_Control	Advisory Output	ISOLATED	Dosage protocol generated post-prediction
Organic_Control	Advisory Output	ISOLATED	Biocontrol guidance generated post-prediction
Soil_Health_Index	Synthetic Redundancy	REMOVED	Collinear handcrafted linear sum of NPK & pH
Rainfall_Category	Discretization	REMOVED	Redundant binned copy of Rainfall

3. Target Class Distributions

Target 1: Pest_Name (25 Classes): Balanced multi-class target representing ~4.0% uniform prevalence across all 25 pest species.
Target 2: Risk_Level (3 Classes): High (63.79%), Medium (34.03%), Low (2.18%). Mitigated via balanced cost-sensitive class weights.